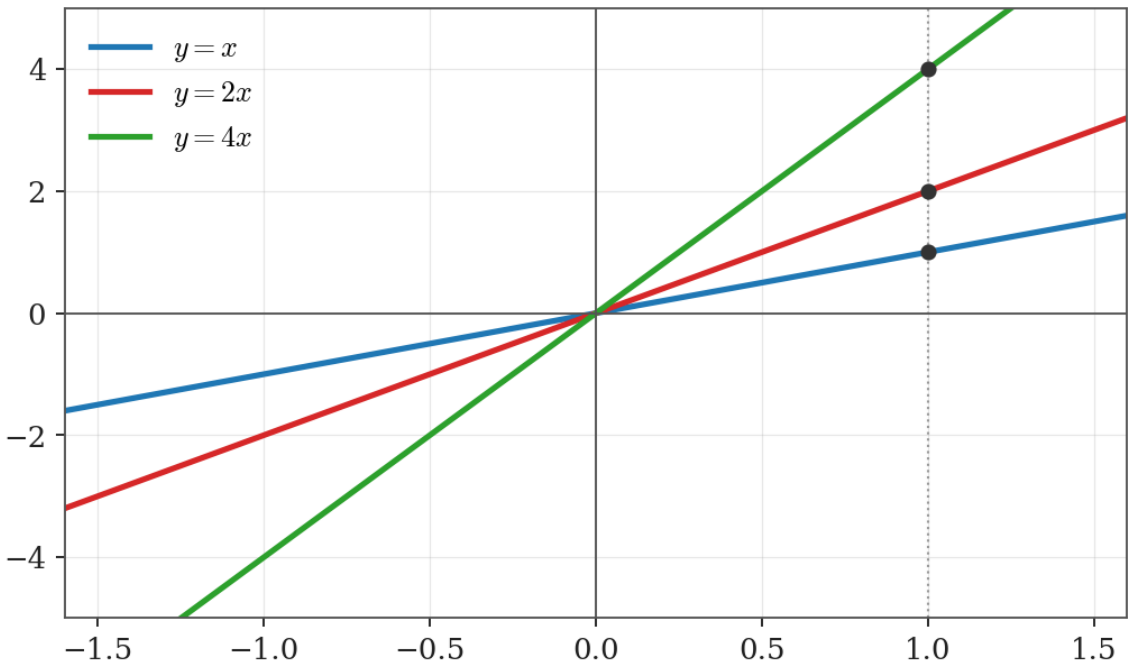
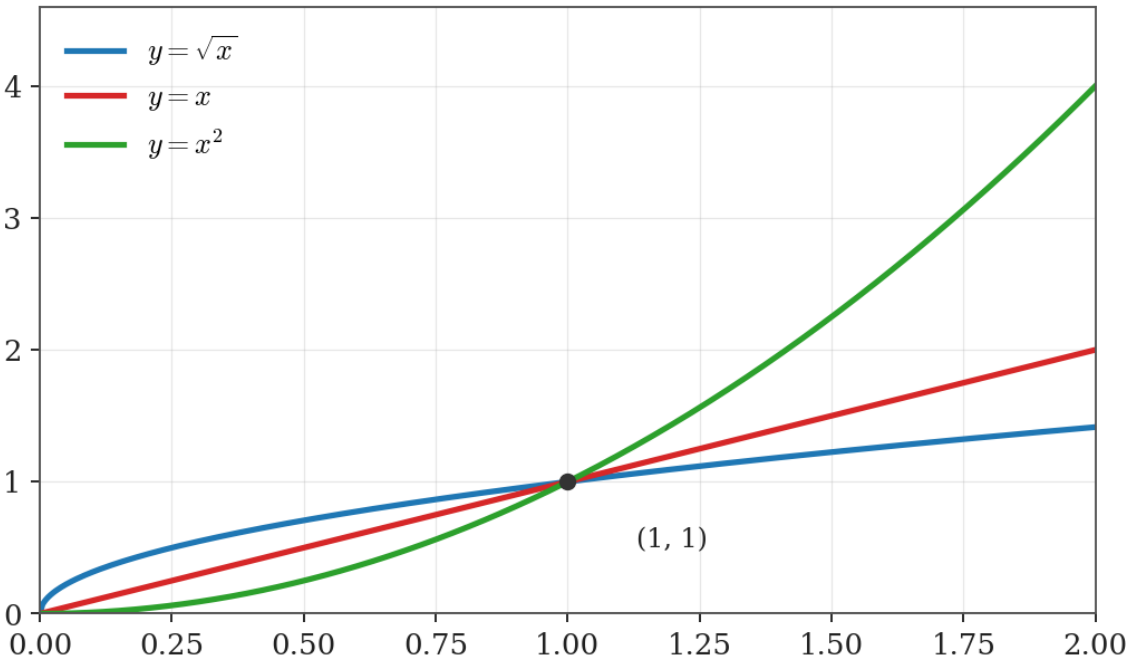
NMRC Math Exercises 2026: Solutions

Exercise 1 — Function plots

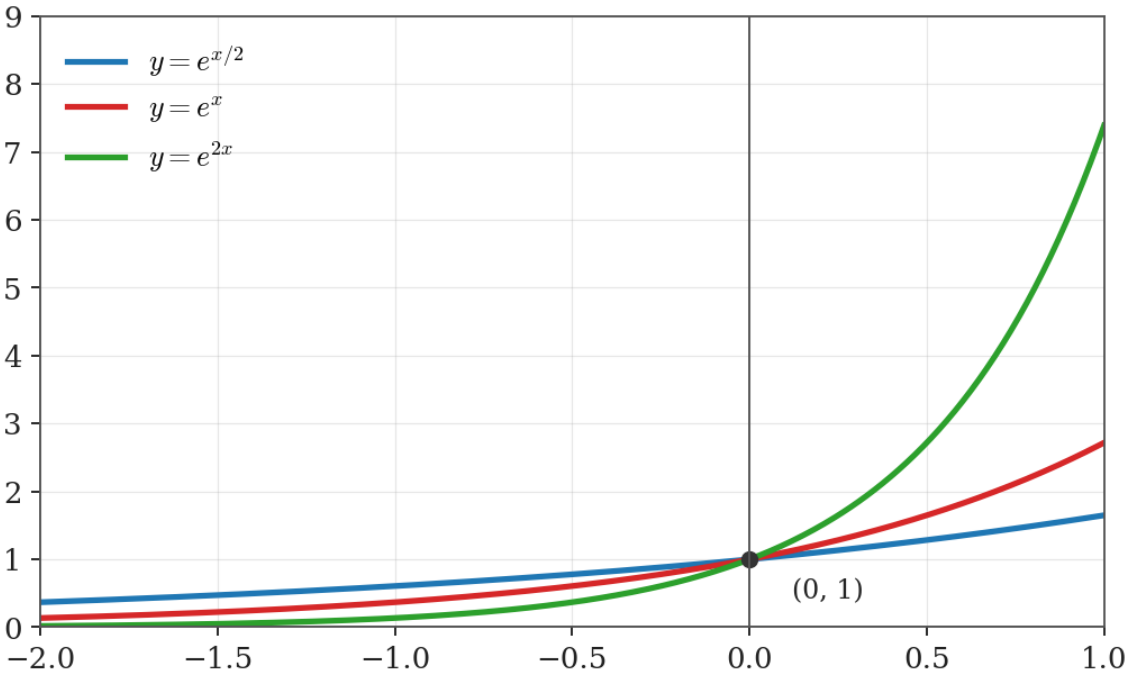
a) Plot of $y = x$, $y = 2x$ and $y = 4x$:



b) Plot of $y = \sqrt{x}$, $y = x$ and $y = x^2$:



c) Plot of $y = e^{x/2}$, $y = e^{x}$ and $y = e^{2x}$:



Exercise 2 — Derivatives

(4) $\frac{d}{dx} 3x^x$

The variable appears both in the base and in the exponent, so rewrite it with the identity $x^x = e^{x \ln x}$:

$$f(x) = 3x^x = 3e^{x \ln x}$$

Apply the chain rule with outer function $V(u) = 3e^u$ and inner function $u(x) = x \ln x$:

$$u'(x) = x \cdot \frac{1}{x} + 1 \cdot \ln x = 1 + \ln x \quad \text{(product rule)}$$

$$f'(x) = V'(u(x)) \cdot u'(x) = 3e^{x \ln x}(1 + \ln x) = 3x^x(\ln x + 1)$$

(5) $f(z) = 2e^z + \frac{1}{z^2}$

A sum, so both terms are differentiated separately:

$$\frac{d}{dz} 2e^z = 2e^z$$

$$\frac{1}{z^2} = z^{-2} \Rightarrow \frac{d}{dz} z^{-2} = -2z^{-3} \quad \text{(power rule)}$$

$$f'(z) = 2e^z - 2z^{-3} = 2e^z - \frac{2}{z^3}$$

(6) $\frac{d}{d\varphi} e^{4\varphi^2}$

A nested function, so the chain rule applies with outer function $V(u) = e^u$ and inner function $u(\varphi) = 4\varphi^2$:

$$V'(u) = e^u, \qquad u'(\varphi) = \frac{d}{d\varphi} 4\varphi^2 = 8\varphi \quad \text{(power rule)}$$

$$\frac{d}{d\varphi} e^{4\varphi^2} = V'(u(\varphi)) \cdot u'(\varphi) = e^{4\varphi^2} \cdot 8\varphi = 8\varphi e^{4\varphi^2}$$

(7)* $\frac{d}{dx} \cos^2(x)$

Rewrite $\cos^2 x = (\cos x)^2$, i.e. $f(x) = (g(x))^2$ with $g(x) = \cos x$ — a nested function, so the chain rule applies:

$$f'(x) = 2g(x) \cdot g'(x)$$

$$g'(x) = \frac{d}{dx} \cos x = -\sin x$$

$$\frac{d}{dx} \cos^2 x = 2\cos x \cdot (-\sin x) = -2 \sin x \cos x$$

which simplifies to $-\sin(2x)$, using the identity $\sin(2x) = 2 \sin x \cos x$.

(8)* $\frac{d}{dx} \left(\frac{x}{1+e^{x^2}} \right)$

A quotient, so the quotient rule applies: for $f = u/v$ it gives $f' = \frac{u'v - uv'}{v^2}$. Here $u = x$ and $v = 1 + e^{x^2}$, with an additional chain rule for the derivative of e^{x^2}:

$$u' = \frac{d}{dx}x = 1$$

$$v' = \frac{d}{dx}(1 + e^{x^2}) = 0 + e^{x^2} \cdot \frac{d}{dx}x^2 = 2xe^{x^2} \quad \text{(chain rule)}$$

$$\frac{d}{dx}\left(\frac{x}{1 + e^{x^2}}\right) = \frac{1 \cdot (1 + e^{x^2}) - x \cdot 2xe^{x^2}}{(1 + e^{x^2})^2}$$

$$= \frac{1 + e^{x^2} - 2x^2e^{x^2}}{(1 + e^{x^2})^2}$$

Exercise 3* — The sigmoid function

For $s(x) = \frac{1}{1+e^{-x}}$, the quotient rule with $u = 1$ and $v = 1 + e^{-x}$ (so $u' = 0$ and $v' = -e^{-x}$ by the chain rule) gives:

$$s'(x) = \frac{0 \cdot (1 + e^{-x}) - 1 \cdot (-e^{-x})}{(1 + e^{-x})^2} = \frac{e^{-x}}{(1 + e^{-x})^2}$$

From here there are various ways to reach $s'(x) = s(x)(1 - s(x))$ — two of them are shown below.

Route 1 — rewrite everything in terms of s. From $s(x) = \frac{1}{1+e^{-x}}$ it follows that:

$$1 + e^{-x} = \frac{1}{s(x)} \quad \Rightarrow \quad (1 + e^{-x})^2 = \frac{1}{s(x)^2}, \qquad e^{-x} = \frac{1 - s(x)}{s(x)}$$

Substituting both expressions into the derivative:

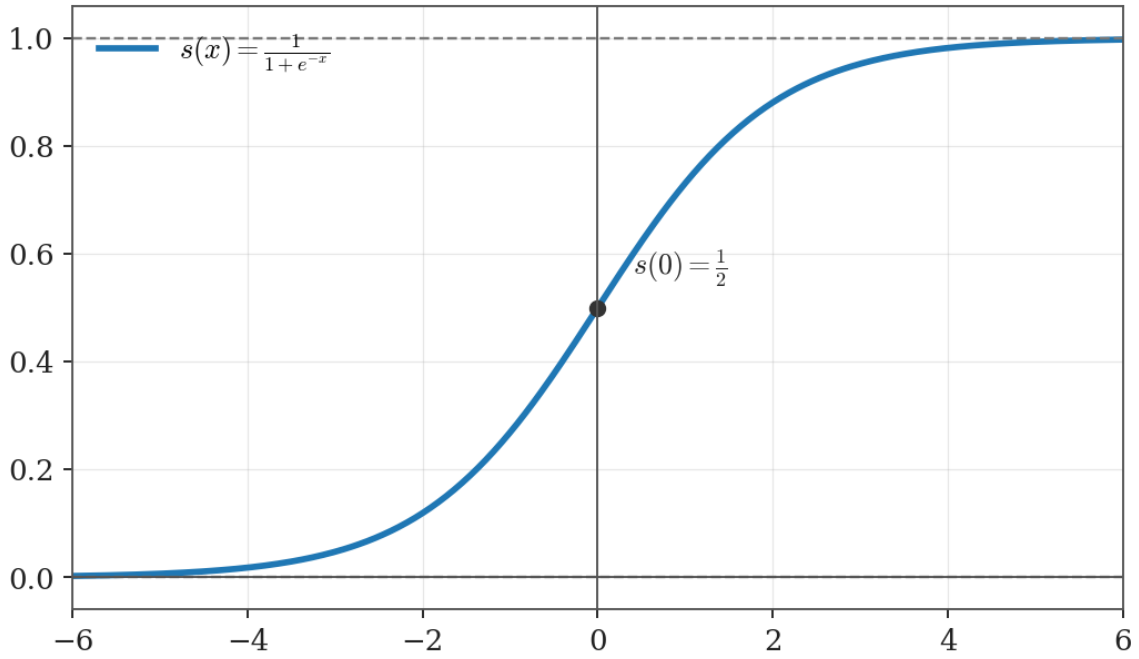
$$s'(x) = \frac{e^{-x}}{(1 + e^{-x})^2} = \frac{1 - s(x)}{s(x)} \cdot s(x)^2 = (1 - s(x)) \cdot s(x) = s(x)(1 - s(x))$$

Route 2 — add $+1 - 1$ (the trick hinted at in the exercise). Adding $+1$ and -1 to the numerator leaves the value unchanged, but groups it around the bracket of the denominator:

$$s'(x) = \frac{e^{-x} + 1 - 1}{(1 + e^{-x})^2} = \frac{(1 + e^{-x}) - 1}{(1 + e^{-x})^2} = \frac{1 + e^{-x}}{(1 + e^{-x})^2} - \frac{1}{(1 + e^{-x})^2}$$

Now both fractions are directly expressible through $s(x)$:

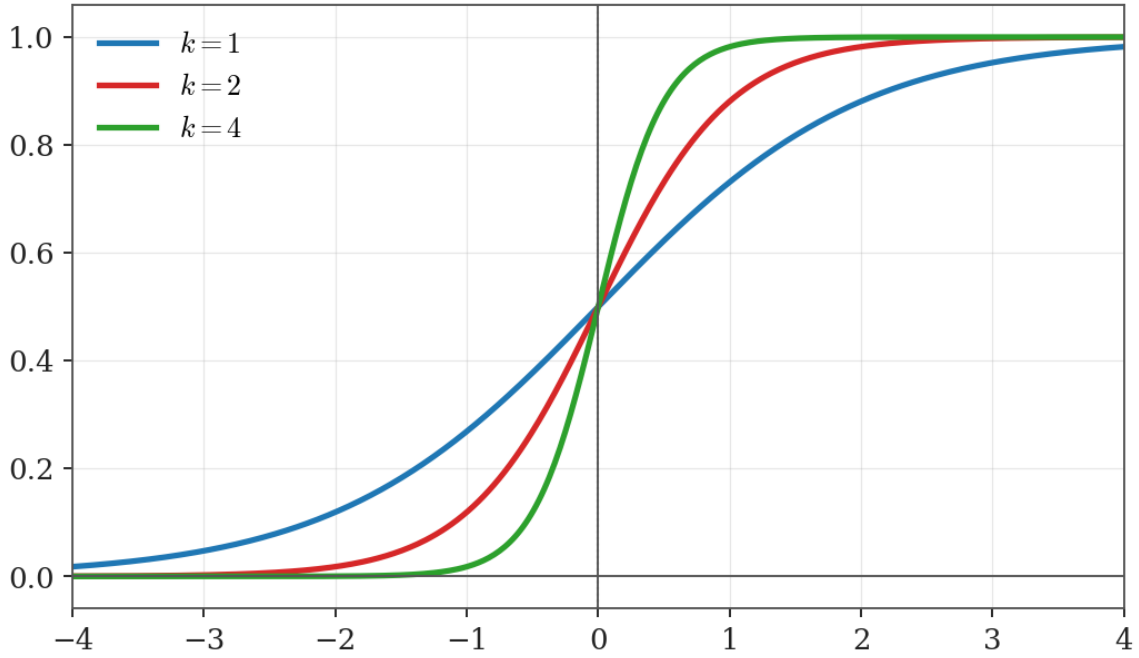
$$s'(x) = \frac{1}{1+e^{-x}} - \left(\frac{1}{1+e^{-x}}\right)^2 = s(x) - s(x)^2 = s(x)(1-s(x))$$



Exercise 4* — A logistic neuron

Given $f(x) = \frac{1}{1+e^{-kx}}$ with $k = 2$. If k increases, then for $x > 0$ the term e^{-kx} becomes smaller, so $f(x)$ moves closer to 1; for $x < 0$ the term e^{-kx} becomes larger, so $f(x)$ moves closer to 0.

A larger k therefore makes the transition at $x = 0$ steeper, so the neuron discriminates more strongly between negative and positive inputs. To obtain this behaviour, choose a larger k, i.e. $k > 2$ (for example $k = 4$):



Exercise 5* — The hyperbolic tangent

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

Apply the quotient rule with $u = e^x - e^{-x}$ and $v = e^x + e^{-x}$:

$$u' = e^x + e^{-x} = v, \qquad v' = e^x - e^{-x} = u$$

$$\frac{d}{dx}\tanh(x) = \frac{u'v - uv'}{v^2} = \frac{v \cdot v - u \cdot u}{v^2} = \frac{v^2 - u^2}{v^2} = 1 - \left(\frac{u}{v}\right)^2$$

Since $\frac{u}{v} = \tanh(x)$:

$$\frac{d}{dx}\tanh(x) = 1 - \tanh^2(x)$$

Exercise 6 — Partial derivatives

(9) $f(x, y) = x^3 + 3xy + 4y^2 + x^2y$

$$\frac{\partial f}{\partial x} = 3x^2 + 3y + 2xy$$

$$\frac{\partial f}{\partial y} = 3x + 8y + x^2$$

(10) $f(x, y, z) = x^2 + 3xyz$

$$\frac{\partial f}{\partial x} = 2x + 3yz, \qquad \frac{\partial f}{\partial y} = 3xz, \qquad \frac{\partial f}{\partial z} = 3xy$$

(11) $f(x, y) = x^y$

For $\partial f/\partial x$ the exponent y is held constant (power rule); for $\partial f/\partial y$ the base x is held constant, so the rule $\frac{d}{dy} a^y = a^y \ln a$ for constant a applies:

$$\frac{\partial f}{\partial x} = y x^{y-1}, \qquad \frac{\partial f}{\partial y} = x^y \ln x$$

Exercise 7 — Gradients

The gradient collects the partial derivatives of a function in a vector.

(13) $f(x, y) = x^2 y$

$$\nabla f(x, y) = \begin{pmatrix} 2xy \\ x^2 \end{pmatrix}$$

(14) $f(x, y, z) = x^2 y \cos z$

$$\nabla f(x, y, z) = \begin{pmatrix} 2xy \cos z \\ x^2 \cos z \\ -x^2 y \sin z \end{pmatrix}$$

(15) $f(x, y, z, t) = x^2 y t^{-1} \cos z$

$$\nabla f(x,y,z,t) = \begin{pmatrix} 2xyt^{-1}\cos z \\ x^2 t^{-1}\cos z \\ -x^2 y t^{-1}\sin z \\ -x^2 y t^{-2}\cos z \end{pmatrix}$$

Exercise 8 — Dot product

$$\mathbf{v} = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}, \qquad \mathbf{u} = \begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix}$$

$$\mathbf{v} \cdot \mathbf{u} = (3\cdot 1) + (2 \cdot 2) + (1 \cdot 4) = 3 + 4 + 4 = 11$$

Exercise 9 — Matrix operations

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 1 & 1 \\ -1 & 0 & 1 \\ -1 & -1 & 0 \end{pmatrix}$$

$$A + B = \begin{pmatrix} 1 & 3 & 4 \\ 3 & 5 & 7 \\ 6 & 7 & 9 \end{pmatrix}$$

$$A + 2B = \begin{pmatrix} 1 & 4 & 5 \\ 2 & 5 & 8 \\ 5 & 6 & 9 \end{pmatrix}$$

$$A^T = \begin{pmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{pmatrix}, \qquad A^T - 2B = \begin{pmatrix} 1 & 2 & 5 \\ 4 & 5 & 6 \\ 5 & 8 & 9 \end{pmatrix}$$